Assignment No 5

Aim: Create a Java program demonstrating single inheritance where a subclass extends a superclass and calls its methods.

Code:

```
import java.util.Scanner;

class Calculation {
    int a, b;
    String original;
    int num;

    public void input() {
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the first number:");
        a = sc.nextInt();

        System.out.println("Enter the second number:");
        b = sc.nextInt();

        System.out.println("Enter a string:");
        original = sc.next();

        System.out.println("Enter a number:");
        num = sc.nextInt();
    }

    public String reverseForString(String str) {
        String rev = "";
        char ch;

        for (int i = 0; i < str.length(); i++) {
            ch = str.charAt(i);
            rev = ch + rev;
        }
        return rev;
    }

    public int reverseForNumber(int num) {

        int rev = 0;
        while(num != 0){
```

```java
            int digit = num % 10;
            rev = rev * 10 + digit;
            num = num / 10;
        }
        return rev;
    }
}

class Arithmetic extends Calculation {

    public void add() {
        System.out.println("Addition : " + (a + b));
    }

    public void sub() {
        System.out.println("Subtraction : " + (a - b));
    }

    public void palindromeForString() {
        String reversed = reverseForString(original);

        if (original.equals(reversed)) {
            System.out.println(original + " is Palindrome");
        } else {
            System.out.println(original + " is Not Palindrome");
        }
    }

    public void palindromeForNumber() {
        int reversed = reverseForNumber(num);

        if (num == reversed) {
            System.out.println(num + " is Palindrome");
        } else {
            System.out.println(num + " is Not Palindrome");
        }
    }
}

public class SingleInheritance {
    public static void main(String[] args) {

        Arithmetic obj1 = new Arithmetic();

        obj1.input();
        obj1.add();
```

```
        obj1.sub();
        obj1.palindromeForString();
        obj1.palindromeForNumber();

    }
}
```

Output:

```
Enter the first number: 12
Enter the second number: 3
Enter a string: madam
Enter a number: 123
Addition : 15
Subtraction : 9
madam is Palindrome
123 is Not Palindrome

Process finished with exit code 0
```